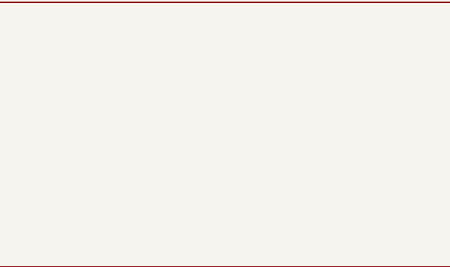


12 V / 100 A THREE-PHASE ALL-NMOS MOTOR DRIVER

DShot150/300/600/1200 via AM32 | STM32G071CBT6 + DRV8323R | 12x N-channel MOSFET, 2 per switch position

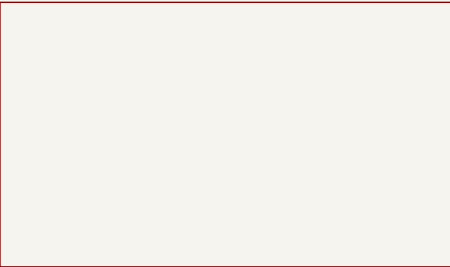
Rev A.1

Power Input



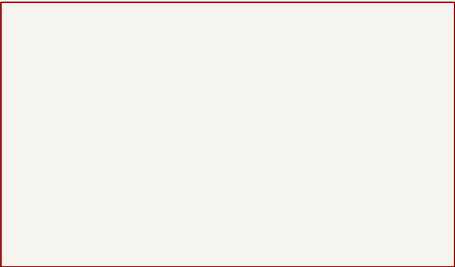
File: 01_power_in.kicad_sch

Gate Driver



File: 02_gate_driver.kicad_sch

Power Stage



File: 03_power_stage.kicad_sch

MCU



File: 04_mcu.kicad_sch

Sense



File: 05_sense.kicad_sch

Protection



File: 06_protection.kicad_sch

SHOOT-THROUGH is prevented in four independent layers:

- 1 TIM1 hardware dead-time generator, DEAD_TIME 40 = 625 ns
- 2 DRV8323R internal cross-conduction lockout + 100 ns
- 3 6x SN74LVC1G97 combinational interlock (sheet 06)
- 4 gate-source pulldowns + per-FET series gate resistors
- 0 nFAULT -> PB12/TIM1_BKIN kills all 6 outputs in hardware

Full budget and derivation: docs/shoot-through.md

100 A continuous / 150 A burst at 12.6 V = ~19 W / ~40 W

dissipated. Needs forced air -- see docs/power-budget.md.

Firmware: AM32, HARDWARE_GROUP_G0_A + NO_PA11_PA12_REMAP.

See firmware/README.md for the target block and the

DRV8323RS SPI init that AM32 does not ship.

STM32G071CBT6 + DRV8323R | 12x TPHP8504PL | DShot via AM32

Top sheet - hierarchy overview

Lucas Poulos

Sheet: /

File: dshot-esc-100a.kicad_sch

Title: 12V 100A Three-Phase All-NMOS Motor Driver

Size: A2

Date:

Rev: A.1

KiCad E.D.A. 10.0.5

Id: 1/7

02 GATE DRIVER — DRV8323R (VQFN—48 RGZ)

DUAL BUILD. RH and RS are the same pinout; only pins 29–32 differ (SLVSDJ3D Table 6–4):

```
pin 29 MODE / SDO    pin 31 VDS / SCLK  
pin 30 IDRIIVE / SDI  pin 32 GAIN / nSCS
```

Populate R203–R206 for RH; fit R207 and route SPI for RS.
NEVER populate both -- see docs/gate-driver-config.md.

The diagram illustrates the circuit for the DRV8323R gate driver. It includes two main ICs: U201A (DRV8323RS) and U201C (DRV8323RS). The power supply section shows a +BATT input connected through several capacitors (C201, C202, C203, C204, C205, C206, C207, C208) to ground. A SW input is connected to a switch and a capacitor (C205). The output section shows three channels (A, B, C) each driving a motor winding through a diode (D201, D202, D203) and a resistor (R201, R202, R203). The control section shows various inputs like INHA, INLA, INHB, INLB, INHC, INLC, ENAB, FB, and nSHDN connected to the appropriate pins of U201C. A PWR_FLAG output is shown at the bottom right.

Buck: 12 V → 3V3 @600 mA. $V_{out} = V_b \cdot (1 + R201/R202)$.
 $R201/R202 = 33.2k/10k$, V_b from SNVSA13 -- see ESC--007.

High-side drive is a CHARGE PUMP, not a bootstrap, so 100% duty cycle is supported and the high-side FET stays fully enhanced at full throttle.

Wired Sheet 2: Gate Driver
Lucas Poulos
Sheet: /Gate Driver/
File: 02_gate_driver.kicad_sch
Title: DRV8323R Smart Gate Driver
Size: A2 Date:
KiCad E.D.A. 10.0.5 Id: 2/7 Rev: A.1

02 GATE DRIVER — DRV8323R (VQFN—48 RGZ)

DUAL BUILD. RH and RS are the same pinout; only pins 29–32 differ (SLVSDJ3D Table 6–4):

```
pin 29 MODE / SDO    pin 31 VDS / SCLK  
pin 30 IDRIIVE / SDI  pin 32 GAIN / nSCS
```

Populate R203–R206 for RH; fit R207 and route SPI for RS.
NEVER populate both -- see docs/gate-driver-config.md.

The diagram shows a dual-channel gate driver circuit. Two DRV8323RS ICs, labeled U201A and U201C, are shown in their VQFN-48 packages. U201A's pins are numbered 1 through 20 on its right side and 2 through 7 on its left. U201C's pins are numbered 28 through 37 on its left side and 45 through 48 on its right side. The circuit includes several passive components: C201 (47µF/50V), C202 (1µF/25V), C203 (100nF), C204 (100nF), C205 (100nF/16V), C206 (100k), C207 (10k), C208 (10k), C209 (10k), C210 (100k), C211 (10k), C212 (10k), C213 (10k), C214 (10k), C215 (10k), C216 (10k), C217 (10k), C218 (10k), C219 (10k), C220 (10k), C221 (10k), C222 (10k), C223 (10k), C224 (10k), C225 (10k), C226 (10k), C227 (10k), C228 (10k), C229 (10k), C230 (10k), C231 (10k), C232 (10k), C233 (10k), C234 (10k), C235 (10k), C236 (10k), C237 (10k), C238 (10k), C239 (10k), C240 (10k), C241 (10k), C242 (10k), C243 (10k), C244 (10k), C245 (10k), C246 (10k), C247 (10k), C248 (10k), C249 (10k), C250 (10k). Resistors include R201 (33k), R202 (33k), R203 (10k), R204 (10k), R205 (10k), R206 (10k), R207 (10k), R208 (10k), R209 (10k), R210 (10k), R211 (10k), R212 (10k), R213 (10k), R214 (10k), R215 (10k), R216 (10k), R217 (10k), R218 (10k), R219 (10k), R220 (10k), R221 (10k), R222 (10k), R223 (10k), R224 (10k), R225 (10k), R226 (10k), R227 (10k), R228 (10k), R229 (10k), R230 (10k), R231 (10k), R232 (10k), R233 (10k), R234 (10k), R235 (10k), R236 (10k), R237 (10k), R238 (10k), R239 (10k), R240 (10k), R241 (10k), R242 (10k), R243 (10k), R244 (10k), R245 (10k), R246 (10k), R247 (10k), R248 (10k), R249 (10k), R250 (10k). Power sources include +BATT, +3V3, and SW. Other components include SWO, FB, PWR_FLAG, and various ground connections.

High-side drive is a CHARGE PUMP, not a bootstrap, so 100% duty cycle is supported and the high-side FET stays fully enhanced at full throttle.

Buck: 12 V → 3V3 @600 mA. $V_{out} = V_{fb}(1+R201/R202)$.
R201/R202 = 33.2k/10k. V_{fb} from SNVSA13 -- see ESC--007.

Wired Sheet 2: Gate Driver	
Lucas Poulos	
Sheet: /Gate Driver/ File: 02_gate_driver.kicad_sch	
Title: DRV8323R Smart Gate Driver	
Size: A2	Date:
KiCad E.D.A. 10.0.5	Id: 2/7

[illegible]

03 POWER STAGE — 12x N-channel, 2 per switch position

TPHR8504PL 40 V / 0.85 mOhm max. 40 V on a 12.6 V bus is deliberate: a 150 A edge into ~5 nH of loop inductance adds 7.5 V of overshoot and real layouts do worse.

EVERY FET has its OWN series gate resistor. Two gates tied directly together resonate through their package inductance and oscillate — separate resistors de-Q that loop. This is mandatory for paralleled FETs, not optional.

Every FET also has a 10k gate-source pulldown, so a floating gate cannot be turned on through Cgd by drain dv/dt.

Dxxx bypasses the common gate resistor on turn-OFF only, so turn-off is faster than turn-on — this widens effective dead time. Asymmetry is in the safe direction. See shoot-through.md.

RS301-304: 4x 1 mOhm 2512 = 0.25 mOhm, 12 W rating, 2.5 W dissipated at 100 A. Hottest discrete on the board.

Wired		Sheet 3: Power Stage	
Lucas Poulos		Sheet: //Power Stage/	
File: 03_power_stage.kicad_sch		Title: Three-Phase All-NMOS Bridge	
Size: A2	Date:	Rev: A.1	
KiCad E.D.A. 10.0.5		Id: 3/7	

03 POWER STAGE — 12x N-channel, 2 per switch position

TPHR8504PL 40 V / 0.85 mOhm max. 40 V on a 12.6 V bus is deliberate: a 150 A edge into ~5 nH of loop inductance adds 7.5 V of overshoot and real layouts do worse.

The schematic illustrates a three-phase power stage. Each phase (A, B, and C) consists of two parallel MOSFET legs. The top leg of each phase contains a MOSFET (GQ301-GQ310), a series gate resistor (R301-R304), and a drain-source resistor (R305-R308). The bottom leg contains a MOSFET (GQ309-GQ312) and a shunt resistor (RS301-RS304). Diodes (D301-D306) are connected between the gates and drains of the top MOSFETs. The output of each phase is taken from the drain of the top MOSFET. The shunt resistors are connected to ground. The schematic also shows the connection of the MOSFETs to the power supply rails (VDD, GND).

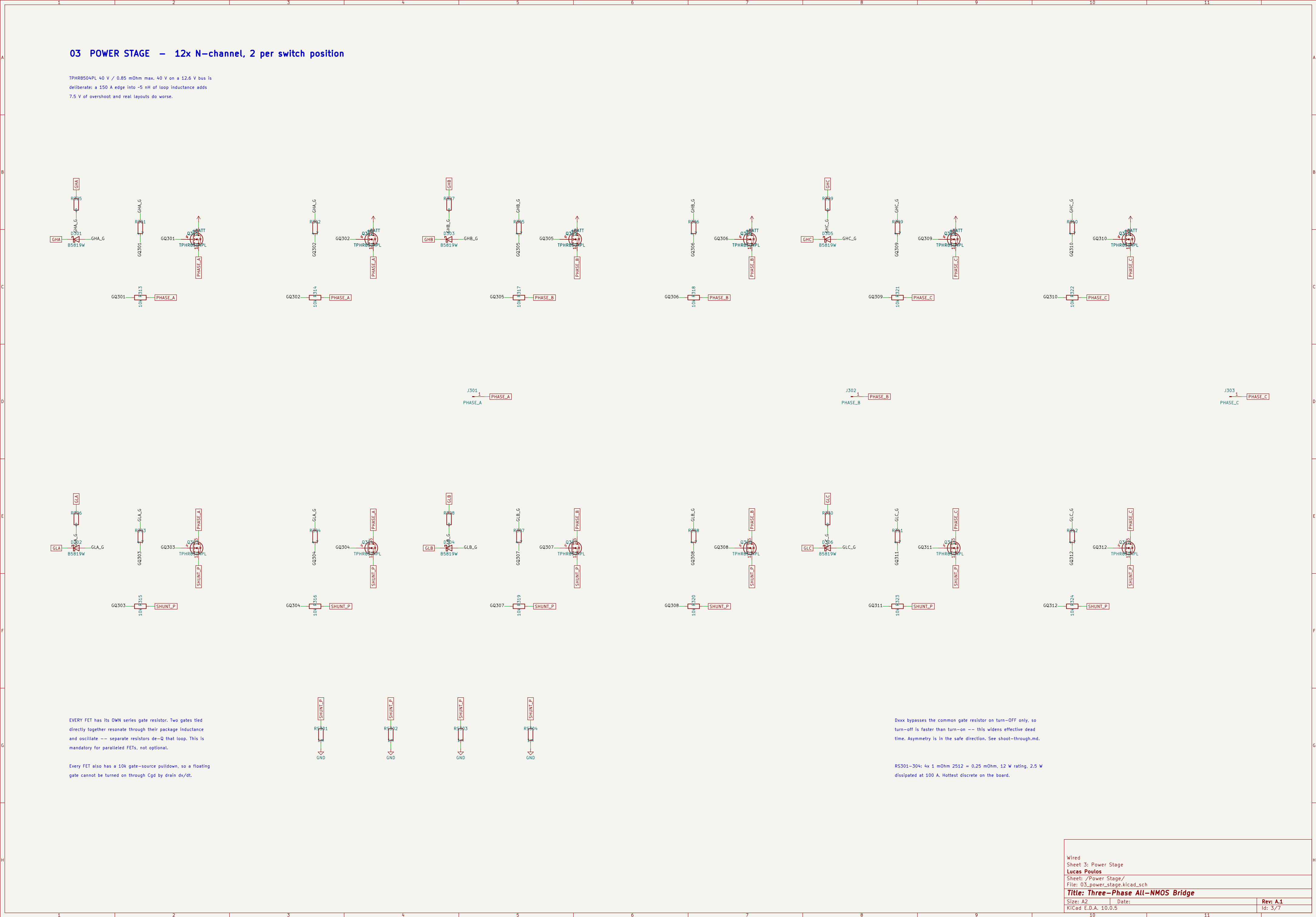
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Size:	A2	Date:	Rev:	A.1
KiCad E.D.A.	10.0.5		Id:	3/7



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Wired		Rev: A.1	
Sheet 3: Power Stage		Id: 3/7	
Lucas Poulos			
Sheet: //Power Stage/			
File: 03_power_stage.kicad_sch			
Title: Three-Phase All-NMOS Bridge			
Size: A2	Date:		
KiCad E.D.A. 10.0.5			

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Size:	A2	Date:	Rev:	A.1
KiCad E.D.A.	10.0.5		Id:	3/7

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Wired		Rev: A.1	
Sheet 3: Power Stage		Id: 3/7	
Lucas Poulos			
Sheet: //Power Stage/			
File: 03_power_stage.kicad_sch			
Title: Three-Phase All-NMOS Bridge			
Size: A2	Date:		
KiCad E.D.A. 10.0.5			

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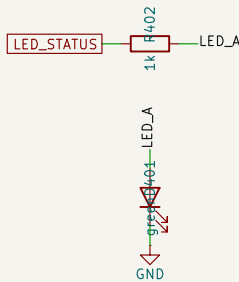
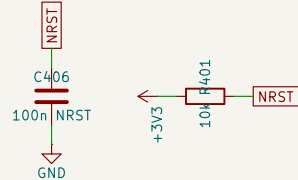
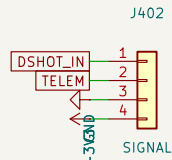
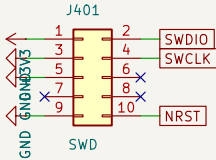
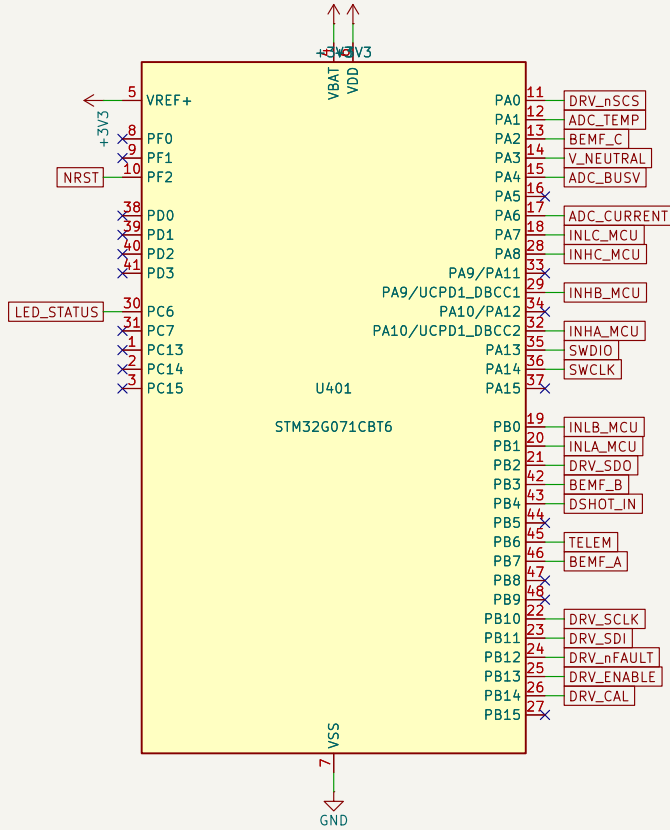
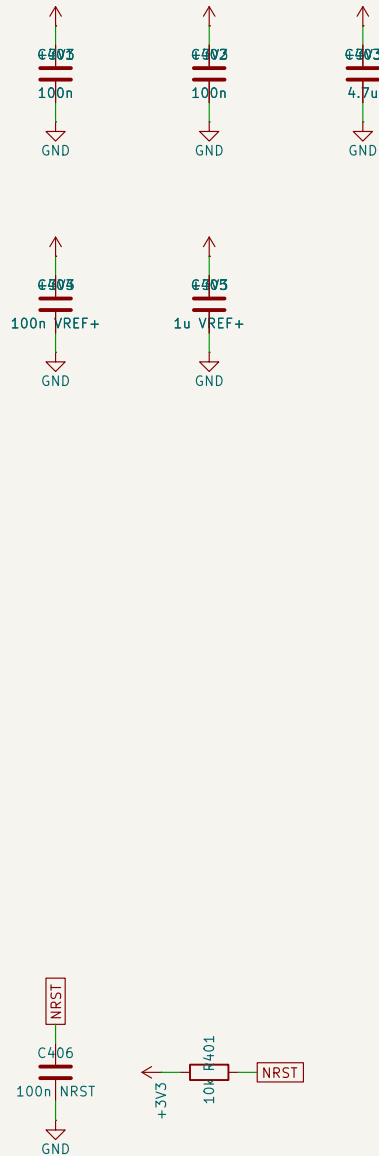
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Wired		Rev: A.1	
Sheet 3: Power Stage		Id: 3/7	
Lucas Poulos			
Sheet: //Power Stage/			
File: 03_power_stage.kicad_sch			
Title: Three-Phase All-NMOS Bridge			
Size: A2	Date:		
KiCad E.D.A. 10.0.5			

04 MCU – STM32G071CBT6 (LQFP–48)

AM32 target: `HARDWARE_GROUP_G0_A + NO_PA11_PA12_REMAP`.
On the 48–pin package PA9/PA10 are DEDICATED pins 29/32;
pins 33/34 are PA11/PA12 and are left unconnected. AM32
remaps by default, which is correct for 28/32–pin parts and
WRONG here –– hence the define. See docs/pinout.md.



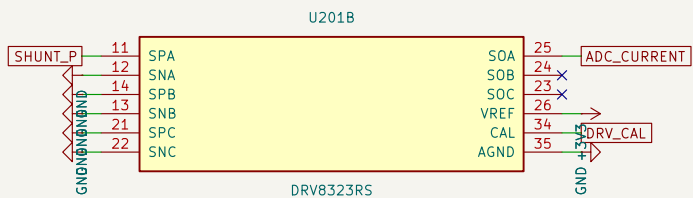
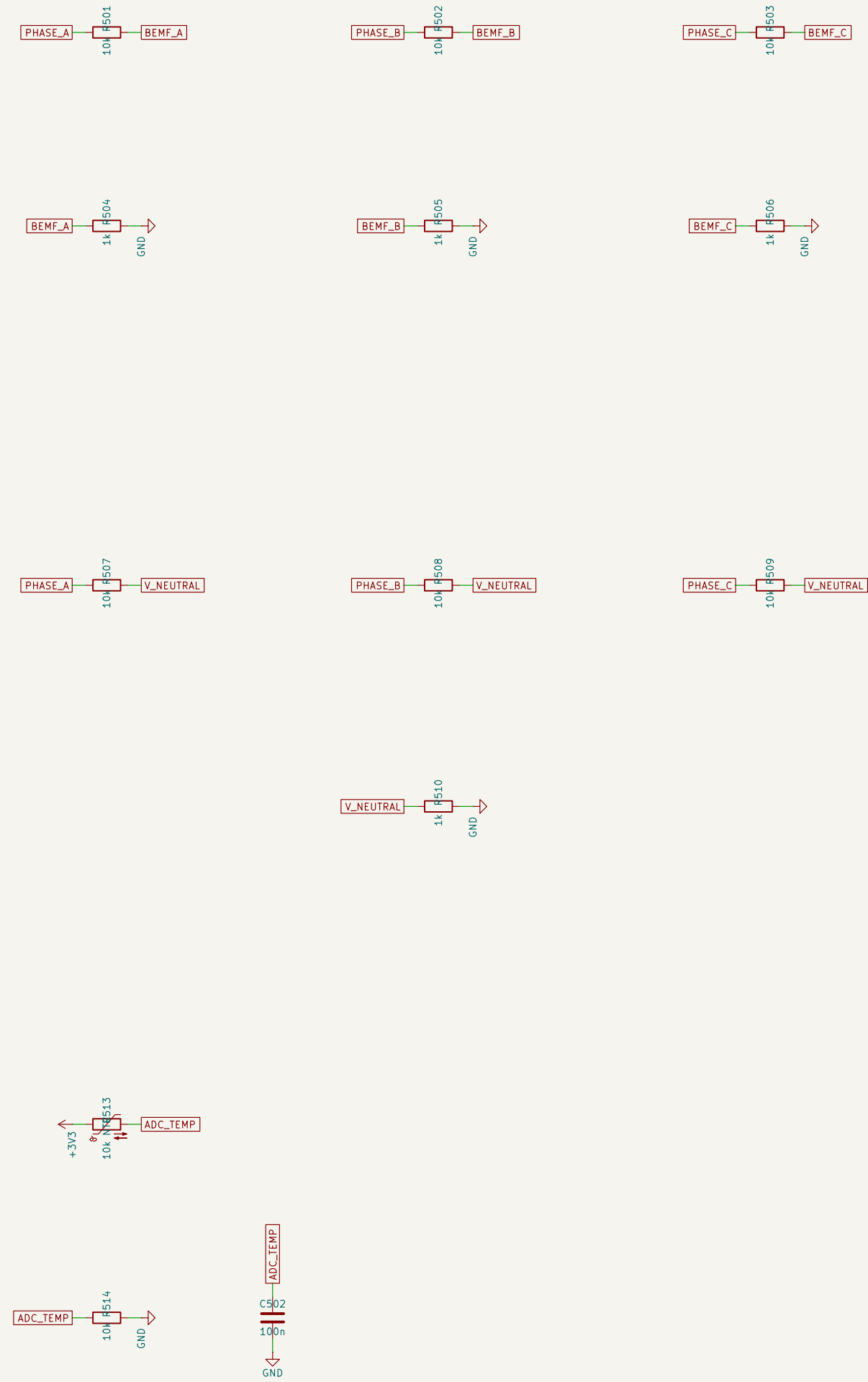
nFAULT lands on PB12 = TIM1_BKIN so a driver fault kills all
six PWM outputs in `HARDWARE`. See `ESC–003`.

Symbol is KiCad's `STM32G081CBTx` copied and renamed: the

G081CB is the same die and bonding as the G071CB (DS12232
covers both) and differs only by the AES block, which has
no pins. Avoids hand–transcribing 48 pins.

05 SENSE – back-EMF, bus voltage, current, temperature

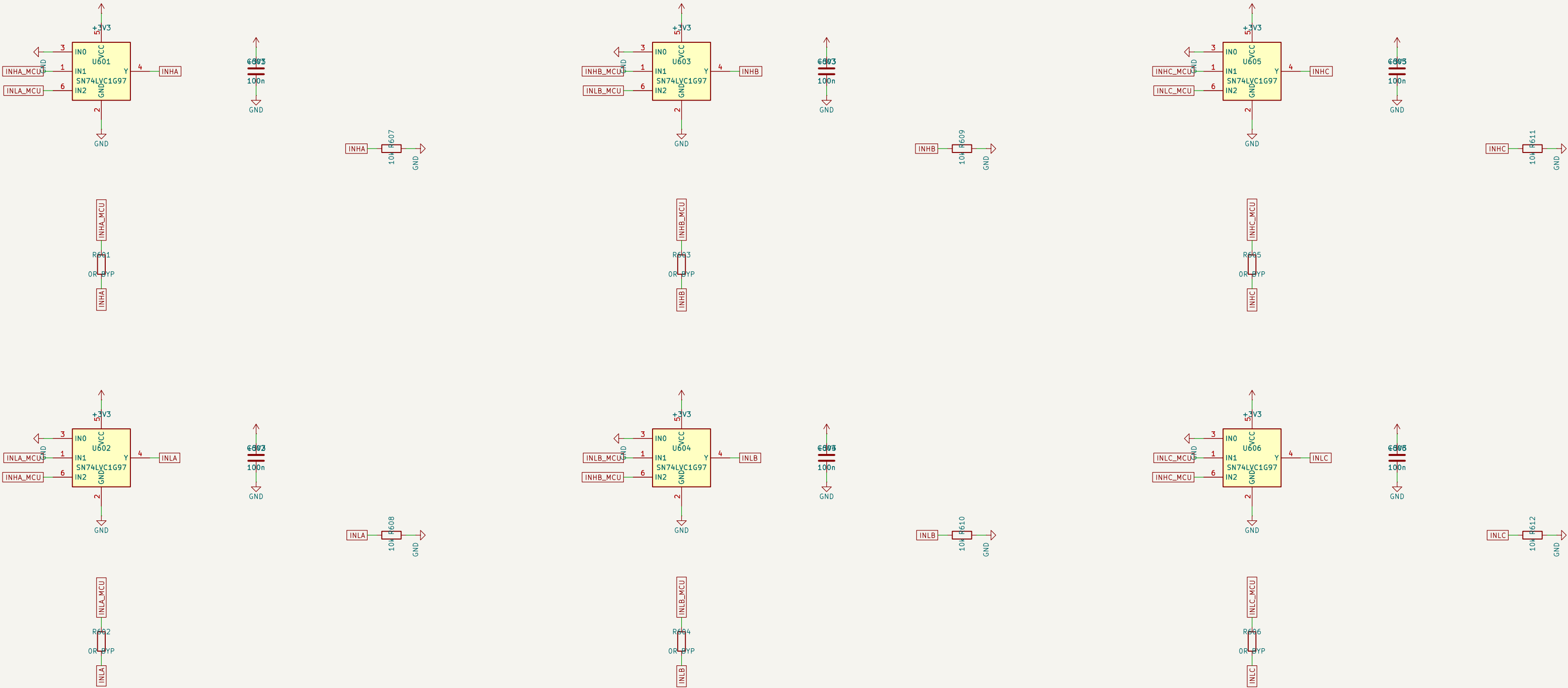
AM32 multiplexes ONE comparator (COMP2) across the three phases. Phase dividers drive COMP2_INM; the virtual neutral drives COMP2_INP on PA3. R507–509 + R510 synthesise the star point, which an unterminated wye or delta motor never brings out. Divider ratio 11:1 matches the phase dividers exactly -- if they differ, zero-crossing detection is biased and timing drifts with throttle.



Only CSA A is used (AM32 reads total bus current). SPB/SNB and SPC/SNC are tied to PGND per SLVSDJ3D Table 6–4: an unused CSA input must be tied, not left floating. Three separate shunts would enable per-phase FOC later at 3x the shunt dissipation -- deferred, not designed out.

06 SHOOT-THROUGH INTERLOCK (layer 3 of 4)

SN74LVC1G97 wired per datasheet SCES416N Figure 6.
'2-Input AND Gate With One Inverted Input':
In0 (pin 3) = GND -> Y = In1 AND NOT In2
high side: In1=INHx_MCU, In2=INLx_MCU
low side: In1=INLx_MCU, In2=INHx_MCU
Both outputs cannot be high at once for ANY input pattern.
This is combinational -- no state to corrupt.



Largely redundant with the DRV8323R's own interlock, and

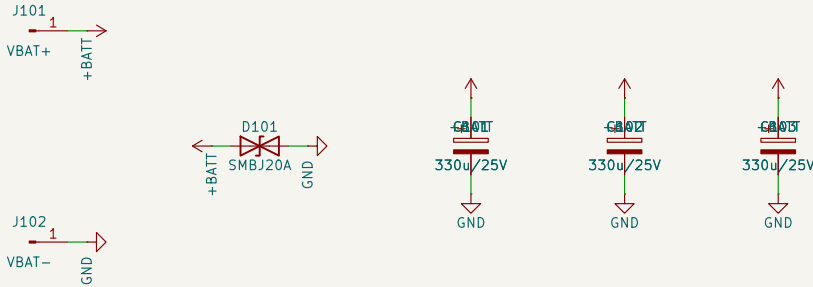
here anyway for the two cases it does not cover: the window before ENABLE+SPI config, and a firmware bug selecting 3x or 1x PWM mode (where INLx stops meaning 'low side on').

R601-606 are OR bypass links for characterisation. Fitting a bypass DEFEATS layer 3 -- do not ship a board with them in.

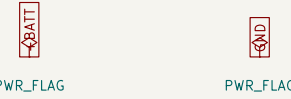
R607-612 pull the driver inputs down, so a tri-stated or unpowered MCU commands 'off' rather than 'undefined'.

01 POWER INPUT + BULK

3S LiPo, 9.0–12.6 V, 100 A continuous / 150 A burst.
D101 SMBJ20A: 20 V standoff clears 4S (16.8 V) and clamps
at 32.4 V, below the 40 V FET rating.



Reverse-polarity protection is deliberately ABSENT: an ideal-diode stage at 100 A costs ~0.5 mOhm and ~5 W plus its own gate drive. Mitigation is the polarised XT90. See ESC-006.



Ripple budget ~50 A RMS at 100 A: C101–103 carry ~8.4 A, the 24x 10uF MLCC array carries the rest. See ESC-001 -- a low-ESR cap at the battery connector on SHORT leads is REQUIRED: lead inductance dominates, not the on-board bank.

